Intracranial Aneurysm Detection in Medical Imaging

Intracranial aneurysms (IAs) are **balloon-like dilations of brain arteries** that arise from weakened vessel walls. They occur in about **3–7% of people globally** [1], and can rupture causing subarachnoid hemorrhage with ~30–50% mortality [2]. Early, accurate detection of aneurysms on routine brain scans (CT or MRI) is thus critical to guiding treatment and preventing life-threatening outcomes [2] [3]. However, most aneurysms are small and asymptomatic, making manual detection laborious. **Radiological imaging modalities** include: - **CT Angiography (CTA):** 3D X-ray scans with contrast that highlight blood vessels, often used clinically for aneurysm screening [4].
- **MR Angiography (MRA):** 3D MRI scans emphasizing blood flow (e.g. Time-of-Flight or TOF-MRA) which can reveal aneurysms without radiation.
- **MRI (T1-post-contrast, T2-weighted):** Structural brain MRI sequences (sometimes with contrast) that may incidentally capture aneurysms.

The current RSNA Kaggle challenge spans **CTA, MRA and conventional MRI (T1 and T2)** [4] [5], mirroring real-world radiology. The dataset is huge – over **6,500 studies and 3,500+ annotated aneurysms** from 18 centers worldwide [6] – making it the largest multi-modal aneurysm imaging collection ever released. Competitors must both **classify** (is an aneurysm present?) and **localize** (where is it?) across 13 vascular sites [7].

Voxels and 3D Imaging

CT and MRI scans are inherently **volumetric** data: each image is a stack of 2D slices forming a 3D grid of **voxels** (volume pixels). A voxel is the 3D analog of a pixel [8], with each voxel's intensity representing tissue density (CT) or signal (MRI) at that location. Voxel *size* (mm^3) depends on in-plane pixel spacing and slice thickness, affecting image resolution and noise [8]. In practice, scans may have anisotropic voxels (thicker slices), requiring interpolation or resampling to train 3D models. Any AI solution must therefore account for 3D data: for example, 3D convolutional neural networks (CNNs) slide 3D kernels through the volume to capture spatial context, unlike 2D CNNs on single slices.

Detection vs. Segmentation

Detection of aneurysms means finding and marking each aneurysm, typically via bounding boxes or coordinates. This contrasts with **segmentation** approaches, where the model outputs a voxelwise mask of suspected aneurysms. Early deep-learning efforts often framed it as segmentation: e.g., training a 3D U-Net to label aneurysm voxels on MRI [9]. More recent work argues for **object-detection networks** (similar to natural-image detectors) because in clinical practice radiologists care about *detecting* lesions, not fine masks [10]. Modern 3D detection frameworks (e.g. `nnDetection` or "Retina U-Net" hybrids) automatically propose candidate regions and classify them. Anchor-based detectors rely on predefined 3D boxes, while **anchor-free methods** (e.g. CPM-Net) directly predict object centers or keypoints [11].

3D CNNs and Transformers

Most models use 3D CNNs (e.g. 3D U-Net or 3D ResNet backbones) to process volumetric scans. For example, Nguyen *et al.* (not cited here) use a 3D squeeze-excitation U-Net. A growing trend is 3D **Vision Transformers** (ViTs), which apply self-attention to 3D patches. Ceballos-Arroyo *et al.* introduced a 3D ViT pre-trained as a **masked autoencoder** (MAE) on large unlabelled CT volumes. They masked out patches near arteries (where aneurysms occur) and taught the model to reconstruct intensities *and* artery-distance maps [12]. This anatomy-guided pre-training improves the model's ability to detect aneurysms after fine-tuning.

Key Technical Concepts

- **Voxel:** The 3D counterpart of a pixel [8]. Medical images consist of voxels in a regular grid, each representing tissue intensity.
- **3D CNN:** Neural network with 3D convolution kernels, capturing volumetric context (depth, height, width). Common for segmentation/detection in CT/MRI.
- **Transformer (3D):** Uses self-attention over 3D patches (blocks of voxels). Can model long-range relationships. Factorized attention (slice-wise then cross-slice) is often used to reduce memory.
- **Segmentation (3D U-Net):** The model outputs a per-voxel label (aneurysm vs. non-aneurysm), often followed by post-processing to find connected components.
- **Object detection:** The model outputs a list of objects with location, size, and confidence. Metrics focus on *lesion-level* sensitivity.
- **Anchor-based vs. Anchor-free:** Anchor-based methods place many candidate boxes of fixed sizes and refine them (e.g. 3D Faster R-CNN). Anchor-free methods predict object centers or use heatmaps, often simplifying hyperparameters [13].
- **Free-Response ROC (FROC):** A performance curve plotting sensitivity vs. average false positives (FP) per scan. In aneurysm detection, it's common to report *sensitivity at 0.5 or 1.0 FP/scan*, reflecting radiologist tolerance [14] [15].
- **Pre-training:** Using large unlabeled data to initialize model weights, then fine-tuning on the aneurysm task. E.g., masked autoencoders improve generalization across sites [12].
- **Multi-modal fusion:** Models may process CTA and MRA (and other MRI) separately or combine channels. For example, *nnDetection* was used on TOF-MRA combined with structural MRI from the ADAM dataset [16].

Recent Advances and Performance

Deep learning for aneurysm detection has rapidly improved. Selected results:

- **CTA (2D detection):** Dai *et al.* (2020) introduced a 2D Faster R-CNN on CTA projections. On 311 patients with 352 aneurysms, they achieved *91.8% sensitivity overall*, and *96.7% for aneurysms >3 mm* [17]. This shows high detection rates, especially for larger aneurysms.
- **3D TOF-MRA (object detection):** Assis *et al.* (2024) proposed a 3D anchor-free detector on TOF-MRA. Compared to segmentation baselines (nnUNet, SCPM-Net), their method reached **86.8% lesion-level sensitivity at 0.53 FP/scan** [9], markedly outperforming previous pipelines at clinically realistic FP rates.
- **3D CT (Transformer with pre-training):** Ceballos-Arroyo *et al.* (2025) used a 3D ViT with anatomy-aware MAE pre-training on thousands of head CTs. On a fine-tuned Chinese CT dataset (Bo *et al.*),

their model achieved *~93% sensitivity at 0.5 FP/scan*, beating strong baselines (e.g. CPM-Net had ~87%) [18]. Notably, on out-of-distribution (external) data, their anatomy-guided pre-training gave a 4–8% absolute gain in sensitivity at 0.5 FP [14].
- **Clinical AI systems:** A recent radiology study reported that state-of-art CNN-based platforms can achieve >90% sensitivity for intracranial aneurysms with FP rates below 2 per scan [19], although combining AI with an expert radiologist yields the best overall accuracy.

These advances underline that **modern models can match or exceed radiologist sensitivity** for aneurysm detection, especially when optimized for low false positives. However, performance varies by aneurysm size and scan type: very tiny aneurysms (<3 mm) remain challenging, and cross-center generalization is an ongoing concern [20] [18].

Strategy for the RSNA Kaggle Challenge

To tackle the RSNA Intracranial Aneurysm Detection task, a systematic pipeline can include:

1. **Data Ingestion & Preprocessing:** Read each 3D scan (CTA, MRA, T1/T2 MRI). Resample volumes to consistent spacing (e.g. isotropic 1 mm voxels) and normalize intensities (e.g. zero-mean/variance). Optionally skull-strip or crop to brain region to remove background. Generate vessel masks or distance maps (as some teams have) to highlight arteries.
2. **Label Handling:** Use the provided annotations (aneurysm centroids or bounding boxes). If only point labels are given, create small 3D cubes or Gaussian blobs around each aneurysm as training targets.
3. **Model Design:**
4. A **3D CNN segmentation network** (e.g. 3D U-Net or nnU-Net) can be trained to output voxel-wise probabilities of aneurysm presence. Post-process by connected-component analysis to yield candidate detections.
5. A **3D object detector** (anchor-free or anchor-based) can be trained to directly predict aneurysm locations. Frameworks like `nnDetection` (which fits a Retina U-Net detector via self-configuring 3D pipelines) can be leveraged [16]. Anchor-free approaches (e.g. predicting object centers or keypoints) reduce hyperparameter tuning [13]. Hybrid models combining CNN and 3D Transformer blocks (inspired by DETR) are also promising.
6. **Multi-modal fusion:** Since scans come in different modalities, one approach is to train separate models per modality and ensemble them, or to concatenate modalities as separate input channels if they are co-registered.
7. **Training:** Use extensive data augmentation (random flips, rotations, intensity shifts) to improve generalization. Given class imbalance (few aneurysms per scan), sample training patches to include a mix of aneurysm-containing and normal regions. Consider **self-supervised pretraining** on all unlabeled scans (e.g. 3D MAE) before fine-tuning on the annotated data [12].
8. **Evaluation:** Monitor lesion-level sensitivity at fixed FP rates (FROC). In development, compute sensitivity vs FP curves on validation sets. Since the leaderboard metric is sensitivity (often at ~0.5–1.0 FP/scan), tune decision thresholds accordingly. Use IoU $\geq$0.3 as the criterion for a true positive [15].
9. **Ensembling and Post-processing:** Combine predictions from multiple models or augmentations to boost sensitivity. Apply false-positive reduction techniques (e.g. another classifier to filter out spurious detections). Because radiologist tolerance is low, ensure the final FP/case is near the target (e.g. $\leq$0.5) while maximizing sensitivity.

10. **Compliance and Computation:** The dataset is large (multicenter 3D imaging), so efficient training on GPUs or TPUs is needed. Use mixed-precision and patch-based training to fit memory (for example, training on sub-volumes of size ~128^3).

By following a structured pipeline—preprocessing, volumetric modeling (3D CNN/Transformer), careful training and FROC-focused evaluation—one can build a robust solution for this task. Leveraging the latest research (e.g. anatomy-aware pretraining [12], anchor-free detectors [13], nnDetection frameworks [16]) should help achieve state-of-the-art sensitivity on this Kaggle challenge.

Sources: Medical imaging definitions and statistics [2] [1]; RSNA/Kaggle challenge details [4] [6]; deep learning methods and metrics [9] [15] [17] [14]. Each cited work provides technical context or empirical results described above.

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